

Modelling Particles, Phase Changes and Mixtures

Molecular Workbench · Concord Consortium

Years 7-10

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

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Simulation: <https://mw.concord.org/modeler/>

Curriculum links: AC9S7U05 · AC9S7U06 · AC9S8U06 · bonding

Learning intentions

- I can describe how particle motion changes as a substance is heated or cooled.
- I can use a particle-level model to explain the difference between a pure substance and a mixture.
- I can relate energy added to a system to a change of state and to bonding between particles.

Getting started

Open Molecular Workbench and load a simple pre-built model from the model library (e.g. a container of particles). Get familiar with the heat/cool or energy controls before you start recording observations.

Tasks

1. Load a starting model at a low temperature. Describe what the particles look like and how they are moving.

2. Slowly add heat or energy to the system. Describe what happens to the speed and spacing of the particles as the temperature rises.

3. Record the particle arrangement (close together / loosely arranged / free-moving) at three different temperature settings.

Temperature setting	Particle arrangement and motion

4. Using your table, explain why heating a substance can cause it to change from a solid to a liquid to a gas, referring to particle motion and spacing.

5. Watch what happens to particles as you cool the system back down. Describe any attraction or bonding you notice between particles at low temperature.

6. If your model shows more than one type of particle, describe one difference you notice between how they behave.

7. A mixture (like salt water) contains more than one type of particle, while a pure substance contains only one type. Using what you saw in the simulation, explain how particle behaviour could help you tell a mixture apart from a pure substance.

Extension (optional)

Explain why understanding particle motion and bonding matters for a real technology, such as designing materials that stay solid at high temperatures or refrigerants that change state easily.
